

Problem

Apply phasor analysis to evaluate the following:

(a) $v = [110 \sin(20t + 30^\circ) + 220 \cos(20t - 90^\circ)] \text{ V}$

(b) $i = [30 \cos(5t + 60^\circ) - 20 \sin(5t + 60^\circ)] \text{ A}$

Step-by-step solution

Step 1 of 2

(a)

Consider the following data:

Given: $v = [110 \sin(20t + 30^\circ) + 220 \cos(20t - 90^\circ)] \text{ V}$

Evaluate voltage expression.

$$\begin{aligned} v &= [110 \sin(20t + 30^\circ) + 220 \cos(20t - 90^\circ)] \text{ V} \\ &= [110 \cos(20t + 30^\circ - 90^\circ) + 220 \cos(20t - 90^\circ)] \text{ V} \\ &= (110 \angle 30^\circ - 90^\circ) + (220 \angle -90^\circ) \\ &= 110 \angle -60^\circ + 220 \angle -90^\circ \\ &= 55 - j95.26 - j220 \\ &= 55 - j315.26 \\ &= 320.02 \angle -80.11^\circ \\ &= 320 \cos(20t - 80.11^\circ) \text{ V} \end{aligned}$$

Therefore, the voltage v is $\boxed{320 \cos(20t - 80.11^\circ) \text{ V}}$.

Step 2 of 2

(b)

Given $i = [30 \cos(5t + 60^\circ) - 20 \sin(5t + 60^\circ)] \text{ A}$

Evaluate the current expression.

$$\begin{aligned} i &= [30 \cos(5t + 60^\circ) - 20 \sin(5t + 60^\circ)] \text{ A} \\ &= [30 \cos(5t + 60^\circ) - 20 \cos(5t + 60^\circ - 90^\circ)] \text{ A} \\ &= 30 \angle 60^\circ - 20 \angle (60^\circ - 90^\circ) \\ &= 15 + j25.98 - 17.32 + j10 \\ &= -2.32 + j35.98 \\ &= 36.05 \angle 93.69^\circ \\ &= 36.05 \cos(5t + 93.69^\circ) \text{ A} \end{aligned}$$

Therefore, the current i is $\boxed{36.05 \cos(5t + 93.69^\circ) \text{ A}}$.